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Charles Mason: an astronomer on difficult ground

Charles Mason was born in Gloucestershire in 1728, a baker's or miller's son. In 1756 James Bradley brought him to Greenwich as an assistant. Not in 1750. The five years sometimes counted backward from 1761 are a way of saying he had already been there a while. The appointment year is 1756.

In September 1760 the Royal Society named him for the transit expedition toward Bencoolen, in Sumatra, with Jeremiah Dixon as his assistant. John Bird, the instrument maker, is the man usually credited with putting Dixon forward. That recommendation is the sturdy tradition. A document of their first handshake is not. This page will not invent the meeting.

The plan did not survive the voyage. After the attack on the Seahorse, told in the transit lesson, they observed from the Cape of Good Hope on 6 June 1761, and only the ending of the transit. Mason's times then entered other people's reductions. James Short compared the Cape with a set of European stations. Mason was an observer in a network, not the owner of the solar parallax.

Pingre later worried that the Cape longitude might be off, and Short had to answer him in print. The argument is a reminder, not a verdict against Mason. A clock time from the Cape is only as good as the place you assign to the Cape. Observers in Europe could not repair a bad longitude by wishing. They could only compare, reject, and say so.

In 1763 the same pair took the Penn-Calvert boundary. A zenith sector watches stars nearly overhead and yields a latitude. Chains and offsets turn that latitude into a mark on the ground. The legal line is geometry. It is useless if you do not know where the instrument stood. That is the transit rule brought ashore: the observer's place is part of the observation.

The survey lesson tells how the tangent was tried, found 22.51 chains off an older point, and corrected without a theatrical restart. Dixon's lesson tells the partnership from his side. Mason's job here is the simpler moral of the instruments. A number in the sky, or a post in the woods, means something only after you can say where you were standing when you took it.

The illustrated lesson is download the PDF.

Sources

- Edwin Danson, on Mason and Dixon's astronomy, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_danson.pdf

- Todd Babcock, on the boundary survey, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_babcock.pdf
- Mason and Dixon's 1767 letter on the line, from the survey record: https://peterschmidt.domains.swarthmore.edu/engl52b/m-d.letter.html

Key facts

- Mason was born in 1728 and became Bradley's assistant at Greenwich in 1756, not 1750.
- He led the 1761 transit party, with Dixon as assistant. Bird is generally credited with recommending Dixon. Their first meeting is not documented.
- After the Seahorse attack they timed the end of the transit from the Cape. Short and others reduced those times with European stations.
- On the boundary, latitude from a zenith sector still required a known place on the ground.